

If X is a scheme (e.g. quasiproj var over a field), then the étale site is the cat of étale maps $U \rightarrow X$ whose morphisms are morphisms of schemes $/X$ and with coverings $(U_i \xrightarrow{q_i} U)_{i \in I}$ s.t. $U = \bigcup_{i \in I} q_i(U_i)$.
Then an abelian presheaf for the étale top is a functor

$$F: X_{\text{ét}}^{\text{op}} \rightarrow \text{Ab}$$

And F is a sheaf if for all covering $(U_i \rightarrow U)_{i \in I}$ in $X_{\text{ét}}$,

$$F(U) \rightarrow \prod_{i \in I} F(U_i) \rightrightarrows \prod_{i,j} F(U_i \times_U U_j)$$

is exact.

Eg For any abelian group Λ ,

$$\underline{\Lambda}(U) = \Lambda^{\pi_0(U)}$$

is an étale sheaf, called the constant sheaf assoc. to Λ .

Eg If $G \rightarrow X$ is a commutative group scheme over X , then

$$U \mapsto G(U) \text{ is an étale sheaf on } X$$

Eg Eg $\cdot \mathbb{G}_a: U \mapsto \mathcal{O}_U(U)$

$$\cdot \mathbb{G}_m: U \mapsto \mathcal{O}_U(U)^\times$$

$$\cdot \mu_n: U \mapsto \{f \in \mathcal{O}_U(U)^\times : f^n = 1\}$$

Eg Let k be a field and let

$$F: (\text{Spec } k)_{\text{ét}} \rightarrow \text{Ab}$$

be a sheaf. Choose a sep closure k^{sep}/k . Let

$$M_F := \varinjlim_{\substack{k'/k \text{ fin} \\ \text{in } k^{\text{sep}}}} F(k')$$

Then M_F is a discrete $\mathbb{Z}[\text{Gal}(k^{\text{sep}}/k)]$ -mod. Conversely, let M be a discrete $\mathbb{Z}[\text{Gal}(k^{\text{sep}}/k)]$ -mod. Then since an étale $U \rightarrow \text{Spec } k$ is

$U = \text{Spec}(\prod_{i \in I} k_i)$ with k_i/k fin sep, we can define

$$F_n(U) = \text{Hom}_{\text{Gal}(k^{\text{sep}}/k)} \left(\text{Hom}_{k\text{-alg}} \left(\prod_{i \in I} k_i, k^{\text{sep}} \right), M \right)$$

Then (exercise)

$M \mapsto F_n$ and $F \mapsto M_F$ defines an equiv of cats

$$\text{AbShv}(\text{Spec } k)_{\text{ét}} \cong \text{Discrete } \mathbb{Z}[\text{Gal}(k^{\text{sep}}/k)]\text{-Mod}$$

Fad $\text{AbShv}(X_{\text{ét}})$ is an abelian cat with enough injectives.

$\Rightarrow$ For any $F \in \text{AbShv}(X_{\text{ét}})$, we get cohomology groups

$$H^i(X_{\text{ét}}, F)$$

by choosing an inj res $F \rightarrow I_0 \rightarrow I_1 \rightarrow \dots$, taking cohom of global sections

$$H^i(X_{\text{ét}}, F) := H^i(I_0(X) \rightarrow I_1(X) \rightarrow \dots)$$

These satisfy

- covariant functoriality in F
- contravariant functoriality in X

Eg If $X = \text{Spec } k$, k a fld, then

$$\text{AbShv}((\text{Spec } k)_{\text{ét}}) \hookrightarrow \text{Disc } \mathbb{Z}[\text{Gal}(k^{\text{sep}}/k)]\text{-Mod}$$

gives isos $H^i((\text{Spec } k)_{\text{ét}}, F) \cong H^i(\text{Gal}(k^{\text{sep}}/k), M_F)$

Thm (Artin) X is a quasiproj var $/\mathbb{C}$ and $X^{\text{an}} = X(\mathbb{C})$ with its analytic complex top, then for any finite ab group Λ ,

$$H^i(X_{\text{ét}}, \underline{\Lambda}) \cong H^i_{\text{sing}}(X(\mathbb{C}), \Lambda)$$

Remark Only works well for ab groups Λ s.t. $\exists n \geq 1$ integer with $n\Lambda = 0$ and such that n is invertible on X , i.e. $n \in \mathcal{O}_X(X)^\times$. So if ℓ is a prime that is invertible on X , we define

$$H^i(X_{\text{ét}}, \mathbb{Z}_\ell) := \varprojlim_n H^i(X_{\text{ét}}, \mathbb{Z}/\ell^n \mathbb{Z})$$

$$H^i(X_{\text{ét}}, \mathbb{Q}_\ell) := H^i(X_{\text{ét}}, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$$

Variant Everything above works best for smooth varieties that are proper over a fld k . For nonproper varieties V over k , there is a variant

$$H^i_c(V_{\text{ét}}, F)$$

called compactly supported étale cohomology with

$$H^0_c(V_{\text{ét}}, F) \subseteq H^0(V_{\text{ét}}, F) = F(V)$$

is the subset of sections whose support in V is proper $/k$.

If V is proper $/k$,

$$H^i_c(V_{\text{ét}}, F) = H^i(V_{\text{ét}}, F)$$

We again define

$$H^i_c(V_{\text{ét}}, \mathbb{Z}_\ell) := \varprojlim_n H^i_c(V_{\text{ét}}, \mathbb{Z}/\ell^n \mathbb{Z})$$

$$H^i_c(V_{\text{ét}}, \mathbb{Q}_\ell) := H^i_c(V_{\text{ét}}, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$$

From now on • Drop subscript "ét"

• We'll work over $\mathbb{F} = \text{alg closure of } \mathbb{F}_q$.

• $V = \text{smooth quasiproj var } / \mathbb{F}$

• $K/\mathbb{Q}_\ell$ a finite ext with ring of ints $\mathcal{O}$, $\ell \neq p$.

- $\Lambda = K, O$, or O/m_0^n .
- Γ = monoid acting on V by proper endomorphisms.

Eg $V = Y$ the Drinfeld curve and

$$\Gamma = (SL_2(\mathbb{F}_q) \times \mu_{q+1}) \rtimes F^N$$

Fact There is a bounded complex $R\Gamma_c(V, \Lambda)$ of $\Lambda\Gamma$ -mods whose cohom groups are $H_c^i(V, \Lambda)$ as $\Lambda\Gamma$ -mods.

Thm (Richard, Requirer) If Γ is a finite group, then

1. $R\Gamma_c(V, \Lambda)$ is chain homotopic to a bounded complex of fin gen $\Lambda\Gamma$ -mods
2. If the stabilizers in Γ of all pts in V have order invertible in Λ , then $R\Gamma_c(V, \Lambda)$ is homotopic to a bdd complex of proj $\Lambda\Gamma$ -mods.